System Software - Utilities

What is Utility Software?

Modern operating systems include a range of programs that perform a collection of useful functions:

- Keep your system secure
- Keep your storage disks organised
- Maintain the system

These programs are called utilities. We shall now take a look at some of these.

Encryption Software

Encryption is where data is scrambled before being sent across a network so that it is unreadable if intercepted. Encryption software facilitates this process. To encrypt data, an encryption key is used which will convert 'plain text' into 'cipher text'. An encryption key is an algorithm which will systematically alter each piece of data in a file. For example, a key may convert each letter in a text file to the next letter in the alphabet:

e.g. 'hello' becomes 'ifmmp'

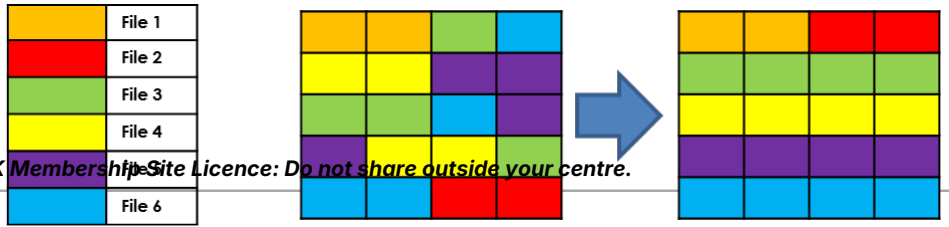
For the 'cipher text' to be converted back to 'plain text', the same key is required by the recipient to reverse the encryption.

Formatting Software (pretext to Defragmentation Software)

Formatting prepares the storage device for data storage. It creates sectors and tracks on which data can be stored. It also creates a special list called the File Allocation Table (FAT). The FAT is used to keep track of where data is located. Each time you write a new file to your hard disk, the table is updated with where the data is stored. Have you ever heard of the police recovering data on a 'wiped hard disk'? How is this possible? When you format your hard disk, you don't actually delete the data on the disk, you simply delete the FAT. The computer can no longer locate any data and thinks it is empty. But in fact the data remains.

Defragmentation Software

As already said, when the disk is formatted, sectors are created to store data. But, these sectors are only 512 bytes in size (tiny!). This means that a file is not stored 'as one' on a hard disk. It is instead stored in 512 bytes sized 'building blocks'. Over time, these 'building blocks' get fragmented over the hard disk. This means that reading a file takes longer as the hard drive needs to be searched in multiple places instead of one area. To help improve read speeds, defragmentation software searches the disk for related data items and reorganises them so they become positioned physically next to one another.



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Data Compression Software

Compression software reduces the size of files. There are various reasons why the size of a file might need to be reduced:

- Less storage space required
- Faster download times – improving online experience
- Faster streaming speeds of video/audio files

There are two ways in which compression software might reduce the size of a file:

Lossy Compression

This is when unrequired data is removed from a file. MP3s are an example of this where sound quality may reduce but not to a point which is noticeable by the listener.

Lossless Compression

This is when data is temporarily removed from the file, but added back (rebuilt) when the file is to be used again. Zip files are an example of this. They will need to be unzipped (extracted) to be useable again.

Backup Software

'Backup software' unsurprisingly allows data to be backed up. A back up is where a copy of some data is made and stored elsewhere. It is vital the data is backed up so that in the event of data loss (through malware, attacks, natural disasters, accidents etc.) the data can be recovered. Imagine if a large online store deleted their database and couldn't recover it – the company would fail overnight! There are two ways in which backup software backs up data:

- Full Backup
- Incremental Backup

Full Backup

With this backup, the software makes a copy of all the files on a system. Doing this each day will take a long time but recovering the data is quick as all the data is stored together, ready to be re-loaded onto the system.

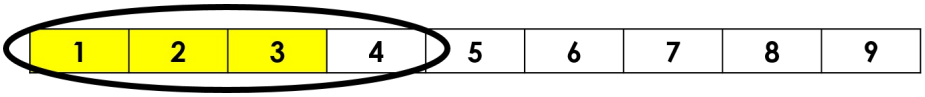
Incremental Backup

With this backup, the software only backs up the files that have changed since the last backup. Instead of a full backup each day the software may backup fully once a week and just do incremental backups the other days. It is much quicker to run the backup, as the system will only back up files that have been changed in the short term. This also leaves an audit trail of the file changes. However, it can take a long time to restore all the incremental backups as well as the last full backup.

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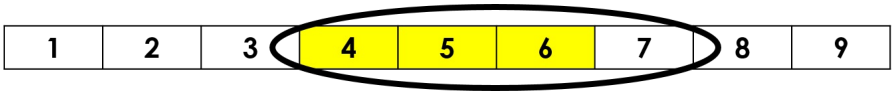
Questions *(The question zone you choose must either match your target grade or be higher!)*

Question Zone 1-3



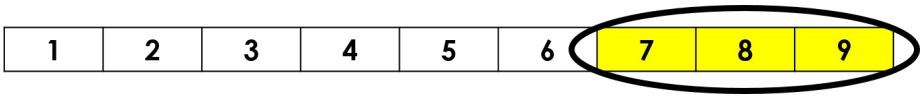
1. Define the term 'Utility Software'. [2]
2. List 5 different examples of 'Utility Software' and for each one give a brief description of its purpose. [5]
3. Explain how encryption software works. [3]

Question Zone 4-6



1. Explain how encryption software works. [3]
2. Explain the need for 'Defragmentation Software'. [4]
3. Describe what compression software does and explain the difference between 'Lossy' & 'Lossless' compression methods, using examples to support your answer. [5]

Question Zone 7-9



1. Describe what compression software does and explain the difference between 'Lossy' & 'Lossless' compression methods, using examples to support your answer. [5]
2. Explain the purpose of 'Backup Software' and describe two methods that 'Backup Software' may use. [6]
3. Explain why a hard drive which has had its data wiped (using formatting software) could still have its original data retrieved. [5]

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New Words / Key Words	Definition and Notes

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Utility 1 –

What:

Use:

Why is it useful?

Utility 4:

What:

Use:

Why is it useful?

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Stick answer sheet here

Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Underlining next & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Encryption: *The process of scrambling (encoding) data before transmission so that if intercepted, it would be unreadable.*

Utility software: *Software which maintains a computer system by keeping it secure, organising data and optimising the system.*

Formatting: *The act of preparing a disk for data storage.*

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words.

Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)

Explain: Explaining is 'saying WHY / HOW something is like that.' (E.G.: A computer will have a CPU so that it can process all of the data it's configured to perform a range of tasks. Primary and Secondary storage is needed because...)

Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)

Describe/Explain/Discuss **using examples:** Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**.

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Stick answer sheet here

Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

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